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AUTOMATIC HEATING AHGR410AW SERIES INSTALLATION AND INSTRUCTION MANUAL

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AHGR410AW DC Inverter Heat Pump

Installation and Instruction Manual

Edition 1.0



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1. Preface

Thank you for purchasing AHGR410AW Series – DC Inverter Heat Pump from Automatic Heating Global Pty Ltd.

In order to provide customers with a high-quality, reliable, and versatile product, this heat pump was produced in accordance with strict design and manufacturing standards. This manual includes all necessary information about installation, handling, maintenance, and troubleshooting. Please read this manual thoroughly before using the unit. The manufacturer will not be held liable in case of damages or injuries resulting from improper installation, handling, maintenance, and troubleshooting that is not in accordance with this manual.

The unit must be installed only by qualified personnel.

The warranty provided is subject to the following conditions:

- The unit must only be installed, opened, or repaired by a qualified installer or an authorized dealer.
- Maintenance and operation must be carried out according to the recommended procedures and frequency stated in this manual.
- Use genuine standard spare parts only.

Failure to comply with these conditions will invalidate the warranty.

AHGR410AW Series – DC Inverter Heat Pump is an efficient, energy-saving, and ecofriendly heat pump that is suitable for space heating and domestic hot water production.

By providing hot/warm water, it can work with any kind of indoor units including fan coils, radiators, and floor heating pipes, which can all be connected to a single monobloc unit. This equipment is designed to allow heat recovery via a super heater that could generate hot water for sanitary and other purposes.

This series of DC Inverter Heat Pumps has the following features:

Advanced Control

The micro-processor controller can be used to easily check or set the running parameters of the heat pump. The centralized control system can also be used to control several units via PC.

Smart Appearance and Easy to Install

The heat pump is designed for visual appeal as well as ease of installation. A monobloc unit has a water pump already included, which makes it compact and also very easy to install.

Quiet Operation

This device uses high quality and efficient compressor, fan, and water pump that ensure low noise level during operation.

Efficient Heat Exchange

This equipment uses a specially-designed heat exchanger that is efficient, helping you save more energy.

Reliable

This device was built to work even under extreme working conditions. It can work under an ambient temperature as low as -15°C for heating.

2. Safety Precautions

To prevent accidents and avoid damage to property and personal injury, please read and understand this manual carefully.

2.1 Signs and Symbols used in this Manual

Warning

Incorrect operation may lead to death or serious injuries.

Prohibited

The action identified is prohibited.

Attention

Incorrect operation may cause minor harm to people or material damages and/or losses.

Mandatory

The action identified is mandatory.

2.2 Warning

Installation

Installer must be professionally trained.

The heat pump must be installed by a qualified and professionally-trained personnel to avoid incorrect installation which could lead to water leakage, electrical shock, or fire.

Provide proper grounding.

Make sure that the unit and power connection have good grounding / earthing to avoid the risk of electrical shock.

Operation

Keep fingers away from fans.

Do not insert fingers or any other objects into the fans or evaporator of the unit as this could result in injury and/or equipment damage.

Shut down the unit in case of emergency.

If you notice something wrong in the unit or if it gives off a strange smell, shut it down immediately by disconnecting the power supply. Continuing to run the unit may cause an electrical short circuit or fire.

Moving and Repairing

Entrust the moving and re-installation of the unit to qualified personnel.

If the unit needs to be moved and be re-installed, please contact a qualified person or company to carry it out for you. Improper installation may lead to water leakage, electrical shock, personal injury, or fire.

Entrust the repair of the unit to qualified personnel.

If the unit needs to be repaired, please contact a qualified person or company to carry it out for you. Improper installation may lead to water leakage, electrical shock, personal injury, or fire.

Do not repair the equipment on your own.

Unless you are professionally trained and suitably qualified, do not attempt to repair or work on the unit as this could result in electrical shock, fire, or other accidents.

2.3 Attention

Installation

Keep away from flammable substances.

DO NOT install the unit near flammable gases or substances. Leakage of these may cause fire.

Provide appropriate machine foundation.

Make sure that the unit is securely positioned and that its base is strong enough to prevent it from falling.

Provide a circuit breaker.

Provide an appropriate and properly-sized circuit breaker for the unit. Lack of circuit breaker could lead to electrical shock or fire.

Operation

Check the baseplate / foundation regularly.

Inspect the baseplate or foundation of the equipment at least once every month to avoid any decline or damage.

Switch off the power during maintenance.

Please switch off the power when cleaning the equipment or during maintenance

Do not use copper or iron as fuse.

It is prohibited to use copper or iron as fuse. The appropriate fuse for the equipment must be selected and installed by an electrician.

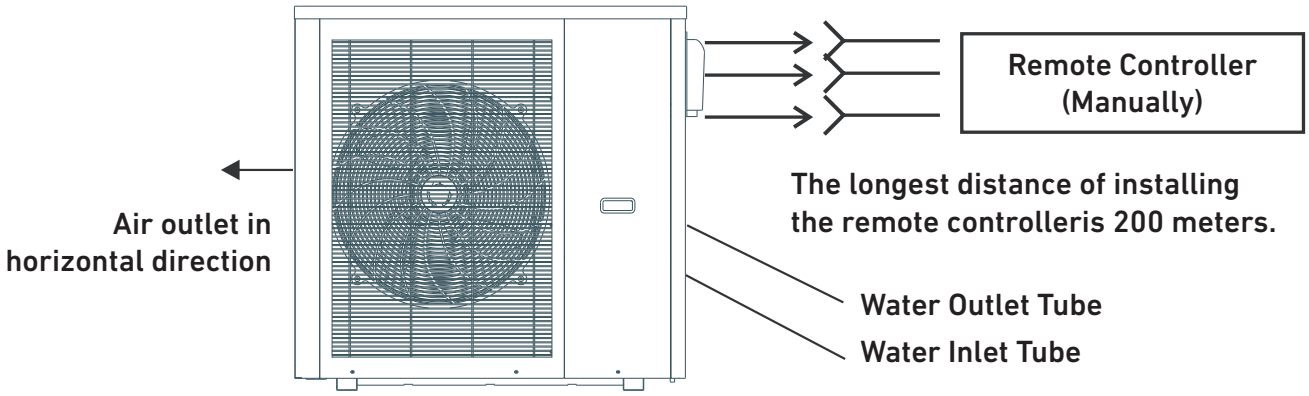
Do not spray flammable gas to the unit.

DO NOT spray flammable gas to the heat pump to avoid fire.

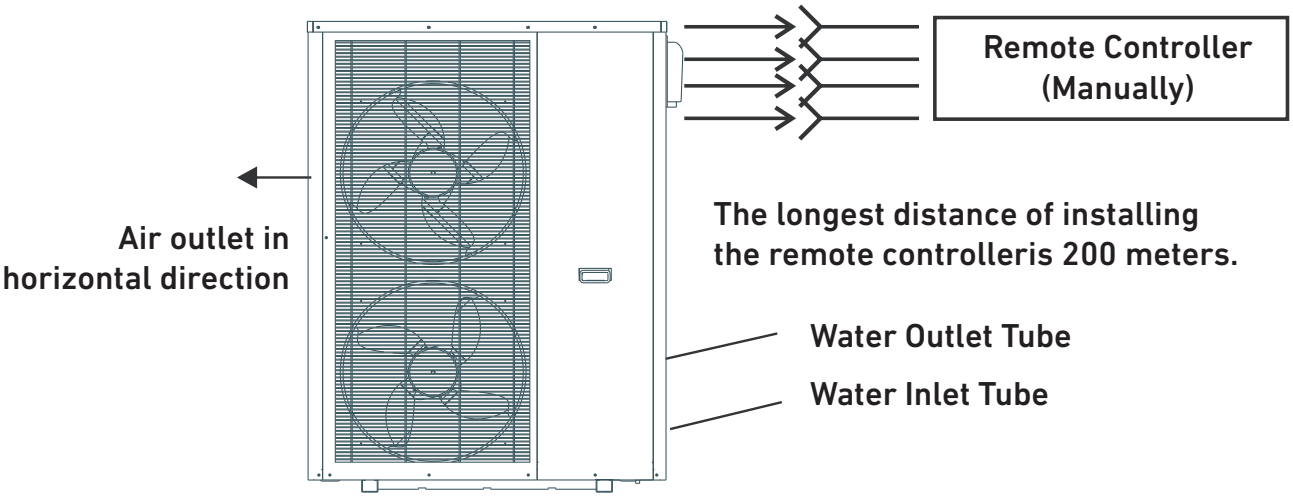
3. General Information

3.1 Heat Pump Appearance and Structure

AHGR410AW6 / AHGR410AW10



AHGR410AW17 / AHGR410AW25



3.2 Equipment Features

Plate Heat Exchangers

The equipment uses SWEP heat exchangers that are compact yet highly efficient.

Eco-friendly Refrigerant

The equipment uses the new generation of eco-friendly refrigerant, R410, which is harmless to the ozone layer.

Wide Operating Temperature Range

The equipment is designed to function normally even at harsh conditions and ambient temperatures as low as -15°C.

The water-cooling system will be opened automatically to guarantee the normal operation of cooling when the ambient temperature is higher than 30°C.

3.3 Specifications

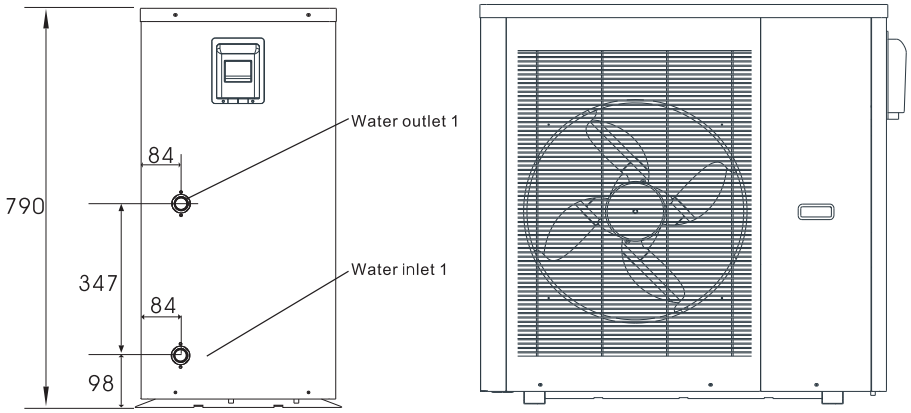
Model		AHGR410AW6	AHGR410AW10	AHGR410AW17	AHGR410AW25
Cooling Capacity	kW	5.4	10	14.5	20
	BTU/h	18426	34121	49489	68260
Heating Capacity	kW	6.4	11.5	16.5	25
	BTU/h	21838	39240	56315	85325
Hot Water Capacity	kW	7.3	13.2	21.6	30
	BTU/h	24909	45040	73702	102364
Cooling Power Input	kW	1.9	3.4	5.6	9.0
Heating Power Input	kW	2.0	3.4	5.1	9.0
Hot Water Power Input	kW	2.1	3.6	6.2	8.2
Operation Electric Current (Max)	A	8.7	15.7	27	14.0
Cooling Capacity Range	kW	1.6~5.4	2.0~10	5.0~14.5	7.0~20.0
Heating Capacity Range	kW	1.9~6.4	2.5~11.5	5.0~16.5	8.0~25.0
Hot Water Capacity Range	kW	3.0~7.3	4.2~13.2	7.4~21.6	11.0~30.0
Cooling Power Input Range	kW	0.6~1.9	1.0~3.4	1.6~5.6	2.5~9.0
Heating Power Input Range	kW	0.6~2.0	0.8~3.4	1.2~5.1	2.5~9.0
Hot Water Power Input Range	kW	0.7~2.1	1.0~3.6	1.5~6.2	2.1~8.2
Power Supply		230V~/50Hz	230V~/50Hz	230V~/50Hz	380V/3N~/50Hz
Compressor Quantity		1	1	1	1
Compressor Model		Rotary	Rotary	Rotary	Rotary
Fan Quantity		1	1	2	2
Fan Power Input	W	75	75	75×2	150×2
Fan Rotational Speed	RPM	850	850	850	900
Noise	dB(A)	54	54	58	62
Water Pump Input	kW	0.05	0.18	0.5	0.65
Water Head	m	6.2	12.5	21	22.5
Water Connection	inch	1	1	1 1/4	1 1/4
Water Flow Volume	m /h3	1.0	1.6	2.8	4.2
Water Pressure Drop (Max)	kPa	10	24	105	71
Unit Net Dimensions (L/W/H)	mm	1008*463*790	953*445*910	996*395*1320	1175*400*1592
Unit Shipping Dimensions (L/W/H)	mm	1060*500*805	1040*490*920	1070*435*1340	1225*430*1600
Net Weight	kg	See nameplate			
Shipping Weight	kg	See package label			

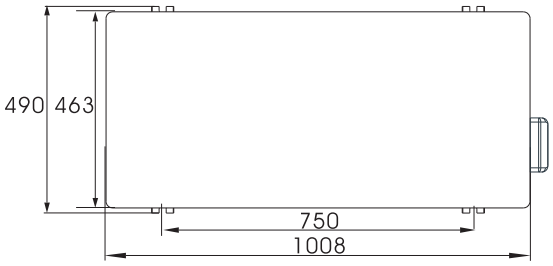
Cooling Working Conditions: (DB/WB) 35°/24°, (Outlet/Inlet) 7°/12°
Heating Working Conditions: (DB/WB) 7°/6°. (Outlet/Inlet) 35°/30°
Hot Water Working Conditions: (DB/WB) 20°/15°. (Outlet/Inlet) 40°/45°.

BS EN 14511-1-2013

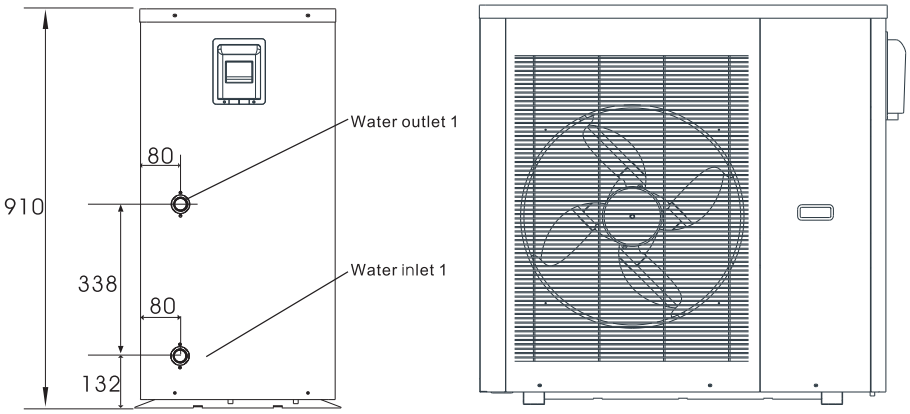
3.4 Drawings and Dimensions

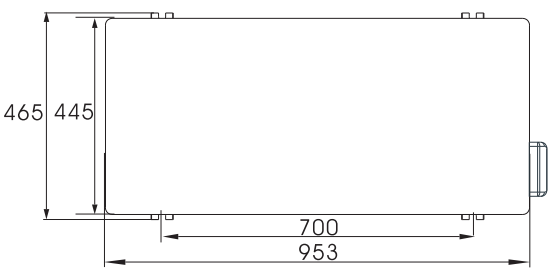
AHGR410AW6





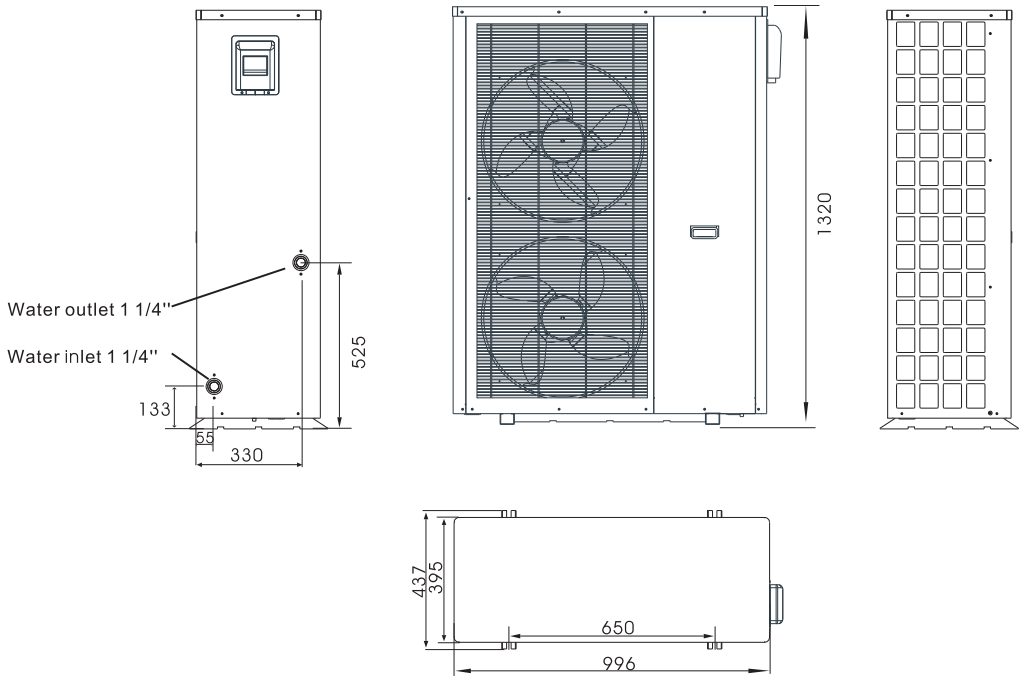
AHGR410AW10



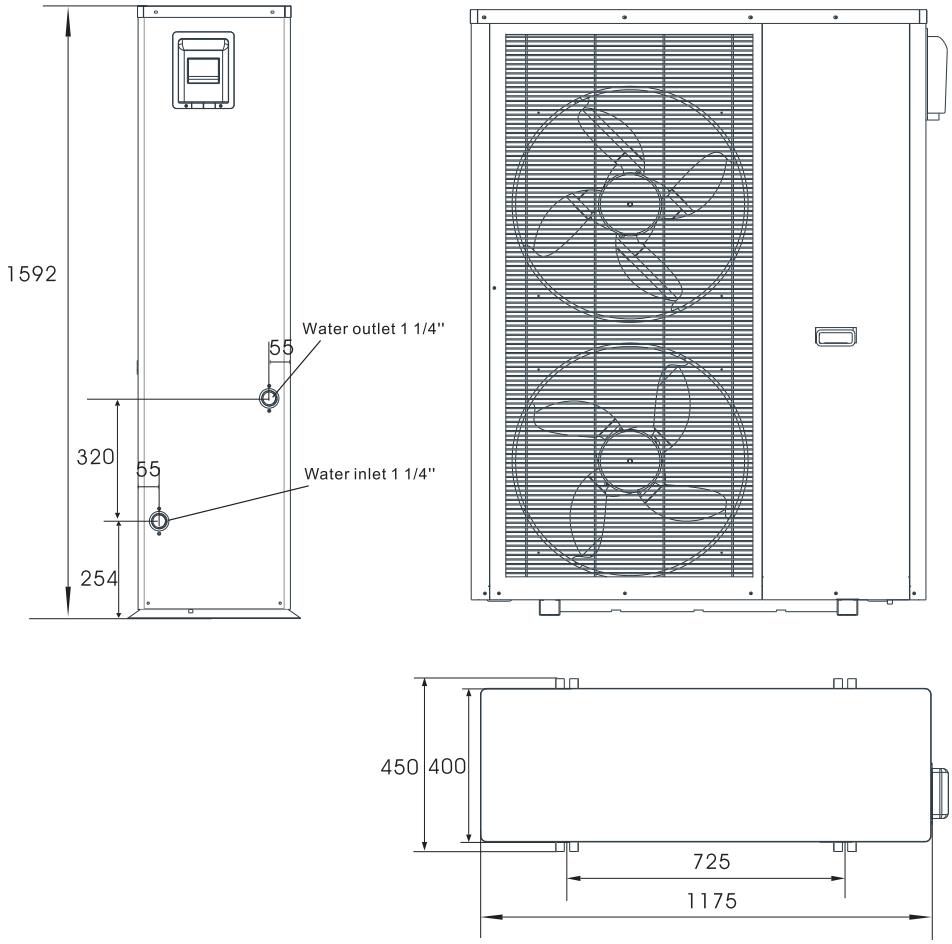


3.4 Drawings and Dimensions (continued)

AHGR410AW17



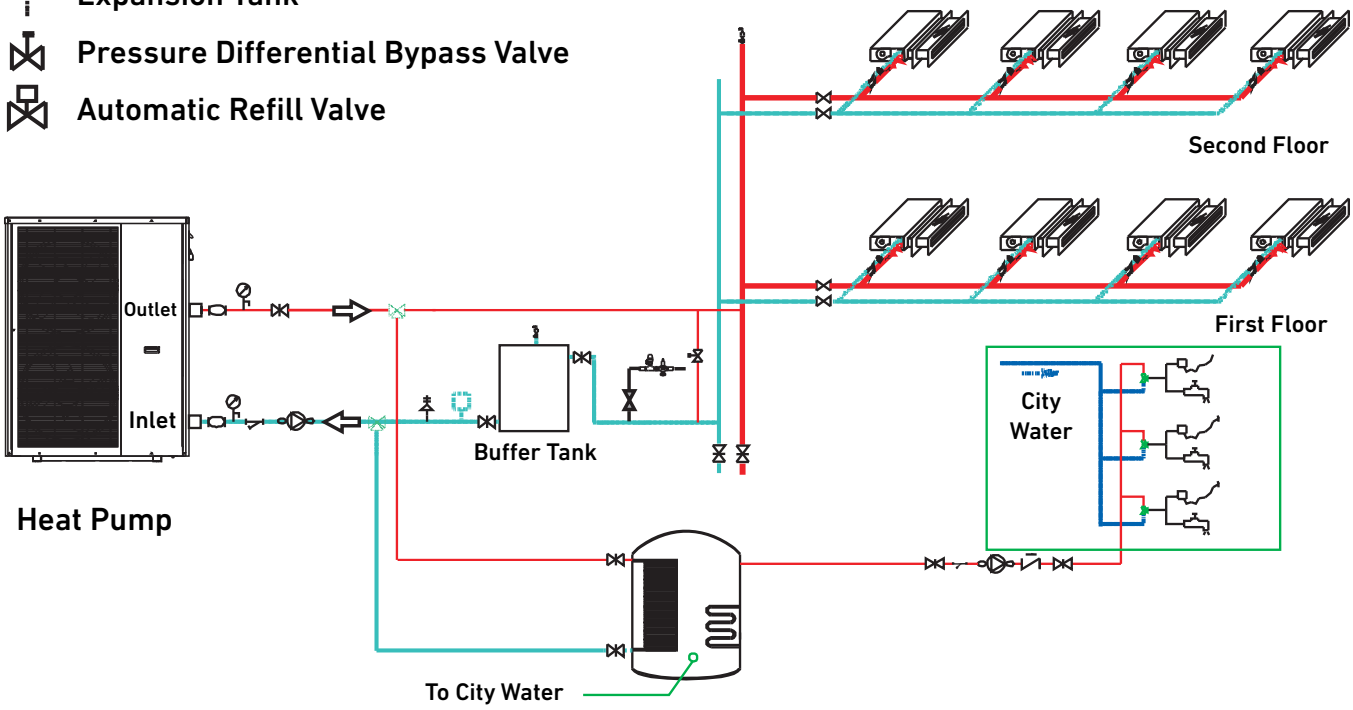
AHGR410AW25



4. Installation

4.1 Schematic Diagram for Air-conditioning

- Connecting Hose
- Water Pump
- Non-Return Valve
- Temperature Meter
- Three-Way Valve
- Expansion Tank
- Pressure Differential Bypass Valve
- Automatic Refill Valve
- General Valve
- Y-Type Filter
- Pressure Meter
- Vent Valve
- Safety Valve



4.2 Equipment Selection & Sizing

Considering local climate conditions, space restrictions, and insulation level, calculate the required cooling (heating) capacity per square meter. Compute the total capacity needed for the whole area.

Then choose the suitable heat pump's model, considering their temperature requirements and outputs specified below:

Cooling Only Unit

Chilled water outlet temperature: 5-15°C
Maximum ambient temperature: 43°C

Heating and Cooling unit

Cooling chilled water outlet temperature: 5-15°C
Maximum ambient temperature: 43°C

Heating unit

Warm water inlet temperature: 40-50°C
Minimum ambient temperature: -10°C

Inverter air source heat pumps are generally used in houses, offices, hotels, and other establishments that need heating or cooling in different areas and that need to be controlled separately.

4.3 Installation Conditions

- The unit can be installed on any area outdoors such as terrace, rooftop, and ground as long as it can carry heavy machinery.
- The location must have good ventilation.
- The place must be free from direct heat and flames.
- A shelter is needed in winter to protect the equipment if exposed to snow.
- There must be no obstacles near the air inlet and outlet of the heat pump.
- There must be an available water channel around the equipment to drain the condensing water.
- The area must be protected from strong winds.
- There must be enough space around the unit for maintenance.

4.4 Machine Foundation

The heat pump can be installed onto a concrete base by expansion screws, or onto a steel frame with rubber feet that can be placed on the ground or rooftop. Make sure that the unit is placed horizontally.

4.5 Water Loop Connection

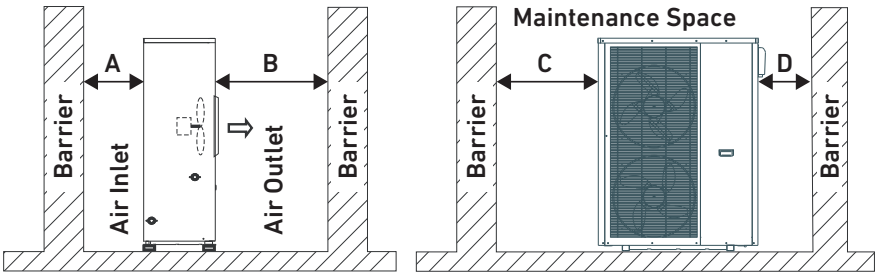
Please consider the following when installing the water pipe connection:

- Try to reduce the resistance on the water through the piping.
- The piping must be clean and free from dirt and debris. Water leakage test must be carried out before insulating to ensure that there will be no water leaking.
- Pipe pressure must be tested separately. DO NOT test it together with the heat pump.
- There must be an expansion tank at the top point of the water loop, and the water level in the tank must be at least 0.5 meters higher than the top point of the water loop.
- The flow switch must be installed inside the heat pump. Ensure that the wiring and action of the switch is normal and controlled by the controller.
- Avoid air staying inside of the water pipe. There must be an air vent at the top point of the water loop.
- There must be a thermometer and a pressure gauge at the water inlet and outlet for easy inspection during operation.

4.6 Power Supply Connection

- Open the front panel, then open the power supply access.
- The power supply must go through the wire access and be connected to the power supply terminals in the controlling box. Then connect the 3-signal wire plugs of the wire controller and main controller.
- If the outside water pump is needed, please insert the power supply wire into the wire access as well and connect to the water pump terminals.
- If an additional auxiliary heater is needed to be controlled by the heat pump controller, the relay (or power) of the auxiliary heater must be connected to the relevant output of the controller.

4.7 Positioning the Unit



The picture shows the location of horizontal air outlet unit.

Attention

Requirements

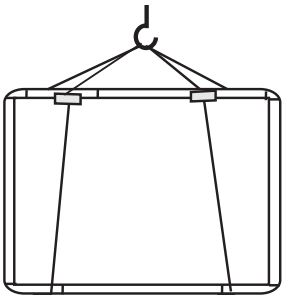
A > 500mm
B > 1500mm
C > 1000mm
D > 500mm

Minimum ventilation distance

4.8 Transit

During transport, please keep the unit in an upright position.

If the unit needs to be hung during installation, an 8-meter cable is needed, and there must be a soft material between the cable and the unit to prevent damage to the heat pump cabinet. *(See picture 1)*



Picture 1

Warning

DO NOT touch the heat exchanger of the heat pump with your bare hands or other objects!

4.9 Trial Run

Inspection before the Trial Run

- Check the indoor unit and make sure that the pipe connection is correctly installed and the relevant valves are open.
- Check the water loop and ensure that the water inside the expansion tank is enough, that the water supply is good, and that the water loop is full of water without any air. Also make sure that the water pipe is well-insulated.
- Check the electrical wiring. Make sure that the power voltage is normal, that the screws are fastened, that the wiring is connected in accordance with the diagram, and that the earthing is connected.
- Check the heat pump unit including all of the screws and parts of the unit to see if they are in good order. Upon powering on, review the indicator on the controller to see if there is any failure indication. The pressure gauge can be connected to the check valve to see the pressure of the system during the trial run.

Trial Run

- Start the heat pump by pressing ⏻ key on the controller. Check if the water pump is running. If it is running normally, there should be a reading of 0.2 MPa on the pressure meter.
- After the water pump runs for 1 minute, the compressor will start. Listen if there is a strange sound from the compressor. If an abnormal sound occurs, please stop the unit immediately and check the compressor. Otherwise if the compressor is running normally, please look at the pressure meter of the refrigerant.
- Then check whether the power input and running current is in line with the manual. If not, please stop and check.
- Adjust the valves on the water loop to make sure that the hot (or cold) water supply through each door is good and meets the requirement of heating (or cooling).
- Check if the outlet water temperature is stable.
- The parameters of the controller are set by the factory. The user is not allowed to change them on their own.

5. Control & Operation

5.1. Main Interface Display and Function

1. Power On Interface



2. Starting Up Interface



Key Functions

	Name	Function
1	On/Off	Switch ON and OFF. Red indicates ON, while grey indicates OFF.
2	Mode	Hot water mode, heating mode, cooling mode, hot water+heating mode, or hot water+cooling mode can be selected
3	Temp. Setting	Sets the target temperature
4	Fast heating	Starts rapid heating. This key will be displayed during heating.
5	Timer Setting	Set the timer. White indicates “disabled”, while green indicates “enabled”.
6	Setup	Check the status of the unit, time, factory parameter, temperature curve, timer setting, and mute setting.
7	Fault	This icon will flash whenever there is an error. The display will enter Failure Record Interface upon tapping this icon.
8	Defrost	Machine is in defrosting mode when this icon is shown
9	Hot Water Mode	Machine is in hot water mode when this icon is shown
10	Cooling Mode	Machine is in cooling mode when this icon is shown

Note: The controller can display the temperature in °F or °C depending on the unit model.

5.2 Usage of Wired Controller

1. On/Off



1. During shutdown, On/Off key will be in gray color. Pressing On/Off key will start up the machine.
2. During startup, On/Off key will be in red color. Pressing On/Off key will shut down the machine.

2. Mode



There are five modes that can be selected in the interface upon clicking the Mode key:

	Mode	Function
1	Hot Water	Advance to hot water mode interface;
2	Heating	Advance to heating mode interface;
3	Cooling	Advance to cooling mode interface;
4	Hot Water + Heating	Advance to hot water + heating mode interface;
5	Hot Water + Cooling	Advance to hot water + cooling mode interface;

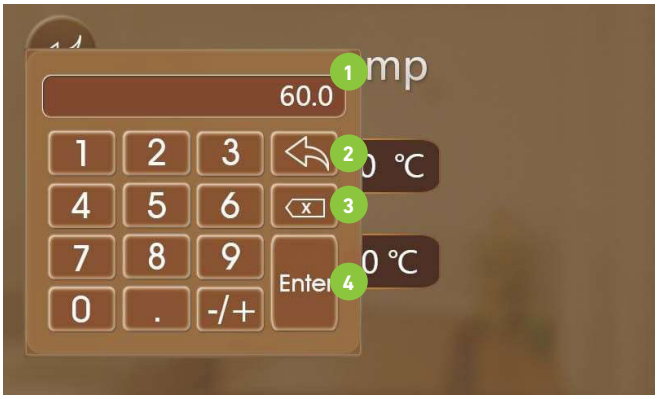
Note: If the unit you have purchased is a heating-only model (without cooling function), cooling mode will not be available and will not be shown on the interface.

3. Temperature Setting



Hot Water + Cooling Mode Example

	Name	Function
1	Back	Goes back to main interface
2	Hot Water Temp	Target temperature of hot water can be set via a pop-up keyboard
3	Cold Water Temp	Target temperature of cold water can be set via a pop-up keyboard



When the target temp. is being set, an onscreen keyboard will pop-up

	Key	Function
1	Current Temp.	Shows current temperature
2	Return	Return back to the main interface
3	Delete	Undo the last action
4	Enter	Save your target temperature and will return back to the main interface

4. Fast Heating

Clicking the fast heating key will lead you to the screen as shown.



	Key	Function
1	Fast Heating	Will trigger the above interface to appear
2	Start	Starts fast heating
3	Stop	Stops fast heating

5. Timer Setting

Clicking the timer setting key, enters the timer setting interface as shown.



	Key	Function
1	Return	Return to the main interface
2	Timed Startup (Green ON / Grey OFF)	Start or turn off the timed start-up function
3	Timed Shutdown (Red ON / Grey OFF)	Start or turn off the timed start-up function
4	Hour Indicator of Timed Startup	Indicates the hour for a timed start-up
5	Minute Indicator of Timed Startup	Indicates the minute for a timed start-up
6	Hour Indicator of Timed Shutdown	Indicates the hour for a timed shutdown
7	Hour Indicator of Timed Shutdown	Indicates the minute for a timed shutdown

In the above picture: Assuming that the unit is operating unmanned, the equipment will start automatically at 17:10 and will shut down at 20:10.

6. Setup

Click the setup key to enter the setup interface



	Key	Function
1	Return	Return to the main interface
2	Operating Mode	View the current operating parameters of the unit
3	Mute	Enable the mute function
4	Factory Parameter	Enter the password to enter the factory parameter settings and status parameters interface
5	System Time Setting	Set the system time
6	Curve	View the temperature curve



Tapping operating mode key (2) will show the above interface;

6. Setup (continued)



Tapping the system time setting key (5) will show the above interface

	Key	Function
1	Return	Return to the main interface
2	Up	Increase the value
3	Down	Decrease the value
4	Cancel	Cancel the current settings and return to the settings page
5	Enter	Enters the current settings

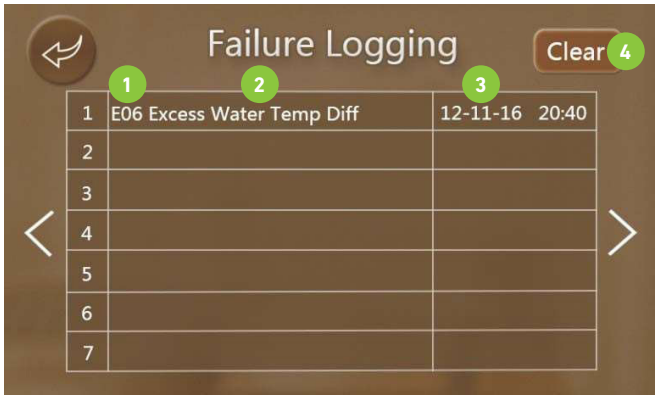


Tapping the mute key (3) will show the above interface;

	Key	Function
1	Auto	Enable auto mode
2	Mute	Enable mute mode
3	Powerful	Enable powerful mode

7. Failure Record Interface

Upon clicking the fault icon on the main interface, the following display will be shown:



	Key	Function
1	Fault Code	Shows fault code
2	Fault Description	Provides info about the fault
3	Fault Date/Time	Indicates when the fault occured: Day-month-year, hour: minute
4	Clear Fault	Clears all fault logs

8. Color Display Calibration

Keep clicking quickly at any blank area on the interface until you hear a long beep. This will trigger the calibration interface to open. Click "+" to start the calibration. Once you hear the beep again, the calibration is finished and will automatically exit.

6. Troubleshooting

6.1 Electronic Control Fault Table

You can refer to the malfunction table below to find out the failure cause & solution

Protect/fault	Fault Code	Reason	Elimination methods
Inlet Temp. Sensor Fault	P01	The temp. sensor is broken or short circuit	Check or change the temp. Sensor
Outlet Temp. Sensor Fault	P02	The temp. sensor is broken or short circuit	Check or change the temp. Sensor
Water Tank Temp. Sensor Fault	P03	The temp. sensor is broken or short circuit	Check or change the temp. sensor
AT Sensor Fault	P04	The temp. sensor is broken or short circuit	Check or change the temp. sensor
Suction Temp. Sensor Fault	P17	The temp. sensor is broken or short circuit	Check or change the temp. sensor
Coil Temp. Sensor Fault	P153	The temp. sensor is broken or short circuit	Check or change the temp. sensor
Exhaust Temp. Sensor Fault	P181	The temp. sensor is broken or short circuit	Check or change the temp. sensor
Exhaust Overtemperature	P182	The temp. sensor is broken or short circuit	Check or change the temp. sensor
Suction Pressure Sensor Fault	PP2	The pressure sensor is broken	Check or change the pressure sensor
Exhaust Pressure Sensor Fault	PP1	The pressure sensor is broken	Check or change the pressure sensor
Low AT protection	TP	The environment temp. is low	
Electric Overheat Protection	E04	The compressor is overloaded	Check whether the system of the compressor is running normally
Communication Fault	E08	Communication failure between wire controller and mainboard	Check the wire connection between remote wire controller and main board
HP Protection	E11	The high-pressure switch is broken	Check the pressure switch and cold circuit
LP Protection	E12	The low-pressure switch is broken	Check the pressure switch and cold circuit
Primary Anti-freezing Protection	E19	The ambient temp. Is low	
Secondary Anti-freezing Protection	E29	The ambient temp. Is low	
Flow Switch Protection	E032	No water / insufficient water in water system	Check the pipe water flow and water pump
Compressor Overcurrent Shutdown Fault	E051	The compressor is overload	Check whether the system of the compressor running normally
Anti-freezing Protection	E171	Water flow is not enough	Check the pipe water flow and check whether the water system is jammed or not
DC Fan Motor 1 Failure	F031	1. Motor is in locked-rotor state 2.The wire connection between DC-fan motor module and fan motor is in bad contact	1.Change fan motor 2.Check the wire connection and make sure they are in good contact
DC Fan Motor 2 Failure	F032	1. Motor is in locked-rotor state 2.The wire connection between DC-fan motor module and fan motor is in bad contact	1.Change fan motor 2.Check the wire connection and make sure they are in good contact
DC Fan Motor Communication Failure	E081	Speed control module and main board communication failure	Check the communication connection

6.2 Frequency Conversion Board Fault Table

You can refer to the malfunction table below to find out the failure cause & solution

Protect/fault	Fault Code	Reason	Elimination methods
IPM Overcurrent Shutdown Fault	F00	IPM input current is large	Check and adjust the current measurement
Compressor Activation Failure	F01	Lack of phase/step or drive hardware damage	Check the measuring voltage Check frequency conversion board hardware
PFC Fault	F03	PFC circuit protection	Check the PFC switch tube if short circuit or not
DC Bus Overload	F05	DC bus voltage > DC bus over-voltage protection value	Check the input voltage measurement
DC Bus Underload	F06	DC bus voltage < DC bus over-voltage protection value	Check the input voltage measurement
AC Input Underload	F07	The input voltage is low, causing the input current to be low	Check the input voltage measurement
AC Input Overload	F08	The input voltage is too high, more than outage protection current RMS	Check the input voltage measurement
Input Voltage Sample Fault	F09	The input voltage sampling fault	Check and adjust the current measurement
Communication Failure between DSP and PFC	F10	DSP and PFC connection fault	Check the communication connection
Communication Fault (DSP)	F11	DSP and main board communication failure	Check the communication connection
Communication Fault (Inverter Board)	F12	Frequency conversion board and main board communication failure	Check the wire connection between remote inverter board and main board
IPM Overheat Stop	F13	The IPM module is overheating	Check and adjust the current measurement
Weak Magnetism Alarm	F14	Compressor magnetic force is not enough	
Input Voltage Lacking Phase	F15	The input voltage underwent phase-loss	Check and measure the voltage adjustment
IPM Current Sample Fault	F16	IPM sampling electricity is faulty	Check and adjust the current measurement
Sensor Fault of Module/Radiator	F17	The transducer is overheating	Check and adjust the current measurement
IGBT Power Device Overheat Alarm	F20	The transducer is overheating	Check and adjust the current measurement
Overload Alarm	F21	Compressor input current is too large	Check and adjust the current measurement
AC Input Overcurrent Alarm	F22	Compressor input current is too large	Check and adjust the current measurement
EEPROM Fault Alarm	F23	MCU error	
Destroyed EEPROM Activation Ban Alarm	F24	MCU error	
LP 15V Underload Fault	F25	The V15V is either overloaded or has low voltage	Check the V15V input voltage if in range 13.5V~16.5V or not
IGBT Power Device Overheat Fault	F26	Transducer temperature is too high	Check and adjust the current measurement

6.3 Parameter List

Meaning	Default	Remarks
Cooling target temperature set point	12°C	Adjustable
Heating target temperature set point	40°C	Adjustable

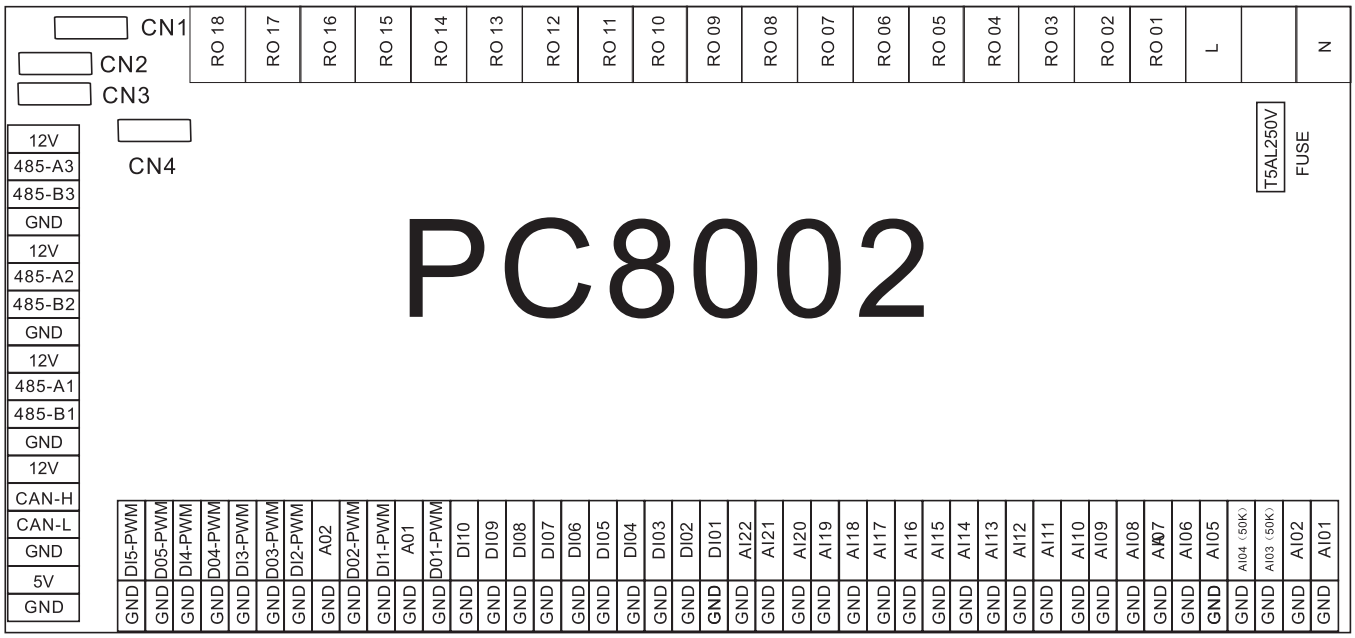
6.4 Equipment Fault Table

You can refer to the malfunction table below to find out the failure cause & solution

Failure	Possible causes for the failure	Solutions
Heat pump cannot be started	• Incompatible power supply • Power supply cable is loose • Circuit breaker is open	• Shut off the power and check the power supply. • Check the power cable and fix the connection. • Check for the cause and replace the fuse or circuit breaker.
Water pump is running with high noise or without water	• Lack of water in the piping • Too much air in the water loop • Water valves are closed • Dirt or blockage on the water filter	• Check the water supply & charge water into the piping. • Discharge the air from the water loop. • Open the valves in the water loop. • Clean the water filter.
Heat pump capacity is low; compressor does not stop	• Lack in refrigerant • Bad insulation on water pipe • Low heat exchange rate on air side exchanger • Lack of water flow	• Check for gas leakage and recharge the refrigerant. • Fix the insulation on the water pipe. • Clean the air side heat exchanger. • Clean the water filter.
High compressor exhaust	• Too much refrigerant • Low heat exchange rate on air side exchanger	• Discharge the redundant gas. • Clean the air side heat exchanger.
Low pressure problem of the system	• Lack of gas • Blockage on filter or capillary • Lack of water flow	• Check for gas leakage and recharge Freon. • Replace filter or capillary. • Clean the water filter and discharge the air from the water loop.
Compressor does not run	• Power supply failure • Compressor contactor is broken • Power cable is loose • Protection on compressor • Wrong setting on return water temp. • Lack of water flow	• Check the power supply. • Replace the compressor contactor. • Tighten the power cable. • Check the compressor exhaust temp. • Reset the return water temp. • Clean the water filter and discharge the air from the water loop.
High noise from the compressor	• Liquid refrigerant goes into the compressor • Compressor failure	• Bad evaporation; check the cause for bad evaporation and get rid of this. • Change the compressor.
Fan does not run	• Failure on fan relay • Fan motor is broken	• Replace fan relay. • Replace fan motor.
The compressor runs but heat pump does not have heating or cooling capacity	• No gas in the heat pump • Heat exchanger is broken • Compressor failure	• Check for system leakage and recharge refrigerant. • Find out the cause and replace the heat exchanger. • Replace compressor.
Low outlet water temperature	• Low water flow rate • Low setting on the desired water temp.	• Clean the water filter & discharge air from the water loop. • Reset the desired water temperature.
Low water flow protection	• Lack of water in the system • Failure on flow switch	• Clean the water filter & discharge air from the water loop. • Replace the flow switch.

7. Main Controller Terminals

7.1 Controller Interface Diagram and Labels



Labels on the main board of the input and output interface:

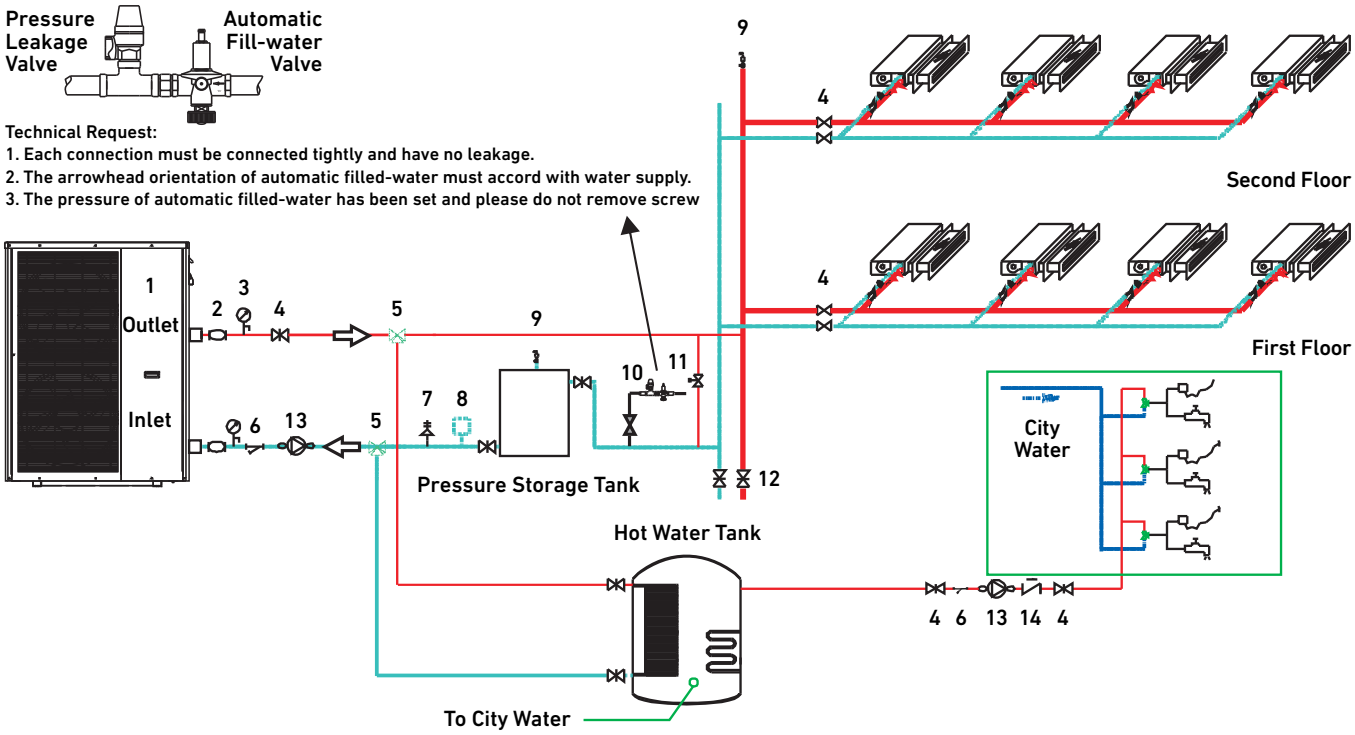
No.	Sign / Label	Meaning
01	RO 01	Compressor (Output 220-230VAC)
02	RO 02	Water Pump (Output 220-230VAC)
03	RO 03	4-way Valve (Output 220-230VAC)
04	RO 04	Fan (Output 220-230VAC)
05	RO 05	Reserved
06	RO 06	Anti-freezing Heater (Output 220-230VAC)
07	RO 07	Crankshaft Heater (Output 220-230VAC)
08	RO 08	Spray Valve (Output 220-230VAC)
09	RO 09	Electric Auxiliary Heater (Output 220-230VAC)
10	RO 10	Reserved
11	RO 11	Alarm (Output 220-230VAC)
12	RO 12	Hot Water Pump

No.	Sign / Label	Meaning
13	RO 13	Electromagnetic Three-way Valve
14	RO 14~RO18	Reserved
15	AC-L	Live Wire (Input 220-230VAC)
16	AC-N	Neutral Wire (Input 220-230VAC)
17	AI01	Water Inlet Temp. (Input)
18	AI02	Water Output Temp. (Input)
19	AI03	System Exhaust Temp. (Input)
20	AI04	Reserved
21	AI05	System Suction Temp. (Input)
22	AI06	System Coil Temp. (Input)
23	AI07	Ambient Temp. (Input)
24	AI08	Water Tank Temp. (Input)
25	AI09-AI20	Reserved
26	AI21	Suction Pressure Input
27	AI22	Exhaust Pressure Input
28	DI01	System High Pressure (Input)
29	DI02	System Low Pressure (Input)

No.	Sign / Label	Meaning
30	DI03	Water Flow Switch (Input)
31	DI04	Emergency Switch (Input)
32	DI05	Mode Switch (Input)
33	DI06	Master-slave Machine Switch (Input)
34	DI07	Electric Heating Overload Protection Switch (Input)
35	485_A1	Frequency conversion board communications
36	485_B1	
37	485_A2	Communication with wire controller and DC fan module
38	485_B2	
39	485_A3	Reserved
40	485_B3	
41	12V	12V(output)
42	5V	5V(output)
43	CN1	Electronic Expansion Valve
44	CN2-CN4	Reserved

8. Appendix

8.1 Appendix 1 – Installation Schematic Diagram (Expandable Water Tank)



Key

1	Main Unit	6	Y-Type Filter	11	Pressure Differential Bypass Valve
2	Connecting Hose	7	Safety Valve	12	Drain Valve
3	Pressure Meter	8	Expansion Tank	13	Non-return Valve
4	General Valve	9	Vent Valve	14	Water Pump
5	Three-way Valve	10	Automatic Refill Valve	15	Fan Coil

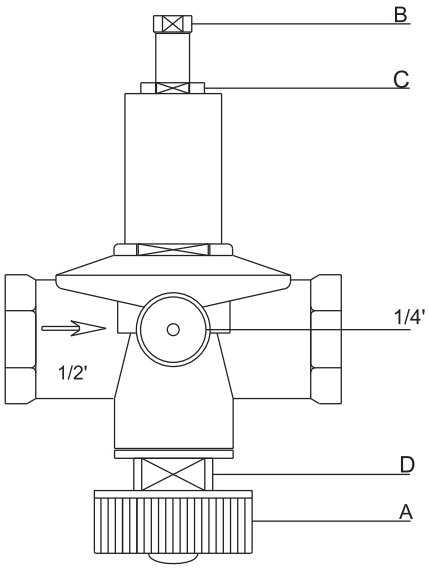
Notes:

1. The supplier only offers the main unit (1). The other parts, which are essential fittings, are provided by the user or the installer.
2. Automatic refill valve (10)is installed at the top point of the water system.
3. The quantity proportion of general valve (4) and three-way valve (5) shall be in accordance with technical regulations, and there should be a three-way valve installed at the farthest point of the water system.

8.2 Appendix 2 – Installation of the Automatic Fill-water Valve

1. When installing the automatic fill-water valve, the arrowhead orientation of inlet water must be in accordance with the orientation of the valve;
2. Automatic fill-water must be adjusted in advance to 1.5 bar;
3. If the pressure of inlet water needs to be re-adjusted, please follow the following procedure:

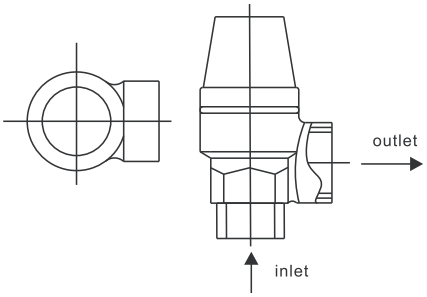
- Open the screw cap (C).
- If the pressure of the water supply needs to be reduced, loosen the screw (B).
- If the pressure of the water supply needs to be increased, tighten the screw (B).
- If the system needs to be filled with water, wrest the handle (A) before filling. Then return and close the handle once the system is full of water.
- The automatic fill-water valve needs to be cleaned regularly. Before cleaning, close the tap, unscrew the plug (D), and remove the inside filter net. Please assemble them again after cleaning



NOTICE: There are two available connections for the pressure meter at the center of the automatic fill-water valve. The screw cap (C) must be tweaked after adjusting the fill-water pressure.

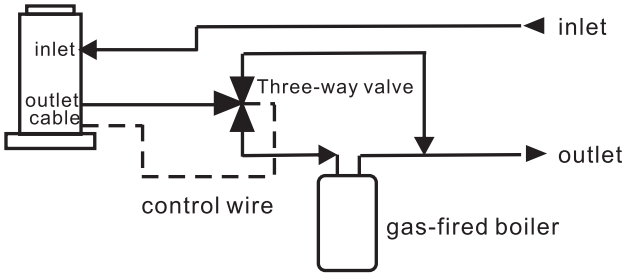
8.3 Appendix 3 – Installation of the Leakage Pressure Valve

1. The action pressure of leakage pressure valve is more than 3bars and the pressure cannot be adjusted.
2. The valve will open automatically to make sure that the water loop of the airconditioning system is safe when the water pressure in the backwater side is higher than the set pressure.

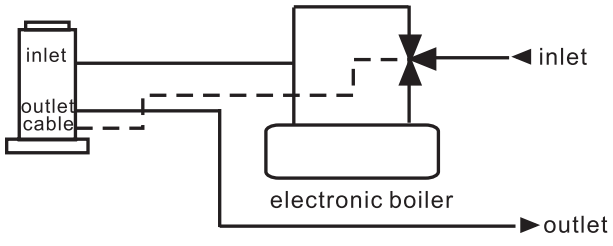


8.4 Appendix 4 – Assistant Heat Sources

The unit allows for the connection of an assistant heat source, which can be a gas-fired boiler or an electronic boiler. The possible schematic diagrams for their connection are as follows:



1. Water Chiller and Assistant Gas-fired Boiler



2. Water Chiller and Assistant Electronic Boiler

8.5 Appendix 5 - Caution & Warning

- The unit can only be repaired by a qualified installing center, personnel, or an authorised dealer. (for European market)
- This appliance is not intended to be used by minors and persons with reduced physical, sensory, and/or mental capabilities, or lack of experience and knowledge, unless they are being supervised by a responsible adult. Keep this equipment away from children.
- Please make sure that the unit and power connection have good earthing to avoid electrical shocks.
- If the supply cord is damaged, it must be replaced by the manufacturer, service agent, or a qualified person in order to avoid hazards.
- Directive 2002/96/EC (WEEE): The symbol depicting a crossed-out waste bin that is underneath the appliance indicates that this product, at the end of its useful life, must be handled separately from domestic waste, must be taken to a recycling center for electric and electronic devices, or handed back to the dealer when purchasing an equivalent appliance.
- Directive 2002/95/EC (RoHs): This product is compliant with directive 2002/95/EC (RoHs) concerning restrictions on the use of harmful substances in electric and electronic devices.
- The unit CANNOT be installed near flammable gas. Gas leakage may cause fire.
- Make sure that there is a circuit breaker for the unit. Lack of circuit breaker could lead to electrical shock or fire.
- The heat pump located inside the unit is equipped with an over-load protection system. It does not allow the unit to start for at least 3 minutes from a previous stoppage.
- The unit can only be repaired by qualified personnel of an installing center or an authorized dealer. (for North American market)
- Installation must be performed in accordance with the NEC/CEC by authorized personnel only. (for North American market)
- USE SUPPLY WIRES SUITABLE FOR A TEMPERATURE OF 75°C.
- Caution: Single wall heat exchanger is not suitable for potable water connection.

8.6 Appendix 6 - Cable specifications

Nameplate max current	Phase line		Earth line	MCB	Creepage protector	Signal line
	Single Phase Unit	Three Phase Unit				
No more than 10A	2x1.5mm²	3x1.5mm²	1.5mm²	20A	30mA less than 0.1 sec	n×0.5mm²
10~16A	2x2.5mm²	3x2.5mm²	2.5mm²	32A		
16~25A	2x4mm²	3x4mm²	4mm²	40A		
25~32A	2x6mm²	3x6mm²	6mm²	40A		
32~40A	2x10mm²	3x10mm²	10mm²	63A		
40~63A	2x16mm²	3x16mm²	16mm²	80A		
63~75A	2x25mm²	3x25mm²	25mm²	100A		
75~101A	2x25mm²	3x25mm²	25mm²	125A		
101~123A	2x35mm²	3x35mm²	35mm²	160A		
123~148A	2x50mm²	3x50mm²	50mm²	225A		
148~186A	2x70mm²	3x70mm²	70mm²	250A		
186~224A	2x95mm²	3x95mm²	95mm²	280A		

If the unit will be installed outdoors, please use the cable which can stand against UV.

AutomaticHeating
efficiency □ performance □ solutions
ABN 73 611 897 600

Tel: 1800 337 959 / Fax: +61 3 9310 5655
sales@automaticheating.com.au
P.O. Box 35, Epping VIC 3076, Australia
www.automaticheating.com.au

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